

# docs-proc-nebius

Serverless document recognition on Nebius GPU Endpoints

Qwen2.5-VL-7B-Instruct · vLLM · single H100 SXM

**#NebiusServerlessChallenge**

## ■ The problem

Identity documents are everywhere — passports, ID cards, residence permits — yet structured extraction is still hard.

- **OCR gives you characters, not meaning.** A `date` could be `DD.MM.YYYY`, `DD MMM YYYY`, or `YYYYMMDD`.
- **The data page can disagree with the MRZ.** Which surname is the right one?
- **A confidence of "high" is useless** if you can't trust it per field.

You want per-field answers you can route on — auto-approve, review, or escalate.

## ■ The solution, in one line

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Take a photo of a document → return **clean, checked JSON fields** — on demand, reproducibly, on **serverless GPU**.

### Your cloud

Images & fields never leave your Nebius endpoint. The model runs on your hardware.

### Trusted output

Describe a type once as a **blueprint**; output is schema-locked JSON with per-field confidence.

### Pay per run

Live endpoint and batch eval share one image & code. Serverless = no idle GPU bill.

## ■ How it's different

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1. **Open model, self-hosted on serverless GPU.** Qwen2.5-VL on a Nebius endpoint — no document leaves your cloud, no GPU bill while idle.
2. **Blueprint-driven, shape-locked output.** Define a type once; the model is **forced** to match it. Always valid JSON — never free text to clean up.
3. **Confidence per field, plus routing.** A score from the model's own token probabilities; weak fields are flagged for review.
4. **Reproducible & portable.** A Dockerfile + open weights → anyone redeploys on any H100 and gets the same result.
5. **One model, live and batch.** Same weights power live recognition and the accuracy job.

# Architecture

Client · browser demo / API / eval job

▼ HTTPS

Nebius Endpoint ingress · TLS, public routing



FastAPI / uvicorn · :8080 (PID 1)

auth · /recognize · blueprints · /metrics

▼ 127.0.0.1:8000

vLLM · Qwen2.5-VL-7B (baked in, offline)

## Container (since v25, minimal)

- **No nginx** — uvicorn serves :8080 directly; Nebius provides the ingress.
- **uvicorn first, vLLM in background** → port opens in seconds, readiness passes while the model warms up.
- **Model baked into the image** ( `HF_HUB_OFFLINE=1` ) → no cold-start download.
- **NOS** (Nebius Object Storage) holds blueprints, uploads, results.

# Blueprints + Guided JSON

A **blueprint** is a JSON schema describing what to extract — per-field instructions and inference hints.

```
"surname": {  
  "inferenceType": "explicit",  
  "instruction": "Surname / last name in uppercase as printed",  
  "required": true  
}
```

- The schema → vLLM's **guided\_json** → constrained decoding **guarantees valid JSON**. No regex, no post-processing.
- Blueprints live in **NOS**, hot-reloaded via **POST /blueprints/reload** — no restart.
- **POST /blueprints/generate** drafts a blueprint from a sample image (two-pass VLM).

# ■ Per-field confidence from log-probabilities

Most systems give one document-level score. Too coarse to route on.

1. Request `logprobs` from vLLM alongside the text
2. Locate the token span for each field value
3. `confidence = round(100 × exp(mean(logprob)))`

**MIDV-2020 calibration:**

**98.2**

mean confidence — correct

**92.7**

mean confidence — wrong

A **5.5 pp** gap — small but directionally correct: wrong extractions do score lower. Good enough for triage routing.

## Five recognition modes + routing

| Mode                      | What it does                             |
|---------------------------|------------------------------------------|
| <code>blueprint</code>    | Extract a named blueprint's fields       |
| <code>auto</code>         | Classify type → pick blueprint → extract |
| <code>raw</code>          | Free VLM text, no parsing                |
| <code>double_check</code> | Extract twice, cross-validate            |
| <code>packet</code>       | Multi-page PDF → per-document            |

### Routing bands

Every response carries a band, so downstream systems decide without inspecting fields:

- `auto_classified` — 85–100
- `review_required` — 50–84
- `escalate_to_operator` — 0–49

Exhaustive & mutually exclusive.



# Live demo



esp\_id · MIDV-2020 (synthetic)

**GET /demo** — zero-install browser UI: upload an image / **PDF** or one-click a bundled sample → fields + confidence.

surname · CORONADO VALVERDE · 97

document\_number · 77748718L · 99

date\_of\_birth · 1978-01-27 · 88

place\_of\_birth · — · 62 → review\_required

# Results & cost on Nebius

## Accuracy — MIDV-2020 (60 docs)

| Field             | Acc. |
|-------------------|------|
| document_number   | 100% |
| nationality / sex | 67%  |
| surname           | 65%  |
| given_names       | 53%  |

Per-type: srb\_passport 77% · esp\_id 68% · grc\_passport 25%\*

(\*polytonic Greek — a 7B limit)

## Performance & cost

| Metric      | Value        |
|-------------|--------------|
| Latency p50 | 1.84 s/doc   |
| Latency p95 | 2.34 s/doc   |
| 60 docs     | 124 s        |
| Cost        | ~\$0.001/doc |
| 1 000 docs  | ~\$1.00      |

Per-second serverless billing — no idle charge.

# Well-Architected on Nebius

Cheap, high-value hardening is implemented; production-scale concerns are called out as future work.

## Operational

One-command deploy, JSON logs, `/metrics`, 35-test smoke.

## Security

Secrets by reference (Mysterybox), app auth, DoS guards, SSRF allowlist.

## Reliability

vLLM self-heal, decoupled readiness, graceful 504s, durable NOS state.

## Performance

7B in bf16 on one H100, baked-in weights, guided JSON + logprobs in one pass.

## Cost

Per-second billing, right-sized 80 GiB disk, one image for live + batch.

## Sustainability

7B not 72B, stop-when-idle, no repeated model transfer.

# Try it

```
docker compose -f docker-compose.cpu.yml up --build # no GPU, mock mode
```

- **Source, blueprints, eval, smoke tests:** [github.com/dsadchikov/docs-proc-nebius](https://github.com/dsadchikov/docs-proc-nebius)
- Deploy to a Nebius H100 endpoint in one command — `scripts/deploy-endpoint.sh`

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